

Your grade: 100%

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Next item →

1. Execute the Coursera Lab Jupyter notebook for this lesson, which implements both EKF and SPKF for joint state and parameter estimation.

1 / 1 point

Which filter gives lower root-mean-squared error on the combined state/parameter estimate?

- ☐ The joint EKF.
- ☒ The joint SPKF.
- ☐ Neither. Both filters give identical RMS error.

✔ **Correct**
Yes, that is correct.

2. For the same Coursera Labs Jupyter notebook, which filter has fewer iterations where the estimation error is outside of the filter's confidence bounds?

1 / 1 point

- ☐ The joint EKF.
- ☒ The joint SPKF.
- ☐ Neither. Both filters have identical number of timesteps where the estimation error is outside of the confidence bounds.

✔ **Correct**
Yes, that is correct.

3. For the same Coursera Labs Jupyter notebook, which filter has narrower confidence bounds? (Which filter has lower root-mean-squared confidence bounds?)

1 / 1 point

- ☐ The joint EKF.
- ☒ The joint SPKF.
- ☐ Neither. Both filters have the same root-mean-squared value for their confidence bounds.

✔ **Correct**
Yes, that is correct.